Requirement Analysis Phase

Script-Controlled ACL - Restrict Record Access Based on Field Value

ServiceNow Platform

Project Title	Script-Controlled ACL - Restrict Record Access Based on Field Value
Platform	ServiceNow (Table, Roles, ACL)
Team Size	5 Members
Lab Duration	3 hrs
Phase	2. Requirement Analysis Phase

Team Members

Member	Name	Role
Member 1	Swetha V	Team Lead
Member 2	Vaitheeshwari S	Requirements
Member 3	Swetha T	Table, Fields, Data
Member 4	Abinesh V	Users, Roles, ACL Scripts
Member 5	Anbuselvam M	Test cases

1. Purpose

This phase defines exactly what the system must do and under which conditions. Requirements are derived from the lab description: only users belonging to the EEE branch may view records where Branch is EEE, and administrators retain full access.

2. Functional Requirements

ID	Requirement	Priority
FR-01	Create a test user (EEE User) with User ID, name and email.	High
FR-02	Create custom roles bb1, bb2, bb3, bb4 and assign them to the EEE User.	High
FR-03	Create custom table Institution Details (u_institution_details) with required fields.	High
FR-04	Insert sample records with branch values ECE, EEE and CSE.	High
FR-05	READ ACL: role bb1 + data condition Branch is EEE + script check; admin has full access.	High
FR-06	CREATE ACL: only role bb2 can create records (New button visible).	Medium
FR-07	WRITE ACL: only role bb3 can edit records.	Medium

FR-08	DELETE ACL: only role bb4 can delete records.	Medium
FR-09	Users without the required role must see no records.	High

3. Non-Functional Requirements

- **Security:** access decisions must be enforced on the server side.
- **Usability:** authorised users should see the list and buttons without extra steps.
- **Maintainability:** ACL script must be short, commented and easy to update.
- **Time:** complete within the 3-hour lab window.

4. Data Requirements - Table: u_institution_details

Field Label	Type	Notes
Student Roll Number	Auto Number (String)	Auto-generated, e.g. INS0001007
Student Name	Reference - User	
Faculty Name	Reference - User	
Branch	Choice	ECE, EEE, CSE
Email	String	
Phone Number	String	
Description	Multi-line String	Used to test WRITE ACL

5. Role Requirements

Role	Purpose
bb1	Read access to EEE branch records (READ ACL)
bb2	Create new records (CREATE ACL)
bb3	Modify records (WRITE ACL)
bb4	Delete records (DELETE ACL)
admin	Full access; bypasses branch restriction

6. Use Cases

Use Case	Actor	Expected Result
View records	User with bb1	Only EEE records displayed
View records	User without role	No records to display
View records	Admin	All records (ECE, EEE, CSE)
Create record	User with bb1 + bb2	New button visible
Edit record	User with bb1 + bb2 + bb3	Fields can be edited
Delete record	User with bb1 to bb4	Delete action available

7. Tools and Environment

- ServiceNow instance (admin access) and security_admin role elevation.

- Modules: User Administration, System Definition > Tables, System Security > Access Control (ACL).
- Impersonate User feature for testing.

8. Assumptions and Constraints

- The team has a ServiceNow developer instance with admin rights.
- security_admin must be elevated before creating ACLs.
- ACLs are created on the table level (u_institution_details), not per field.
- Lab duration is limited to 3 hours.

9. Acceptance Criteria

- bb1 user sees only EEE records.
- User without role sees no records.
- Admin sees all records.
- Create, write and delete work only with bb2, bb3 and bb4 respectively.